

Mohammed Hassan

Embodied and Human-Centered AI

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EDUCATION

The University of Texas at Austin

M.S. in Computer Science

Expected May 2027

B.S. in Computer Science and B.S. in Mathematics

May 2025

PUBLICATIONS AND MANUSCRIPTS

Mohammed Hassan, Zhenyang Chen, Zheng Wang, Zhixin Zhu, Tianlong Chen, Yingyan (Celine) Lin, Chaojian Li. *T3R: Training-Free Two-Stage Token Refinement Towards Efficient and Robust VLA Models*. ASP-DAC 2027. **Accepted**. [Project website](#)

Maintaining anonymity for submission purposes. MAESTRO: Agentic Long-Form Music-to-3D Dance Composition and Editing. CHI 2027. **Under review**. [Project website](#)

Mohammed Hassan, Omar Abouelazm. *Improving BB84 Efficiency with Delayed Measurement via Quantum Memory*. arXiv:2410.21191, 2024. QIP 2025. **Poster accepted**. [Paper](#)

RESEARCH EXPERIENCE

The University of Texas at Austin | VITA Lab

Jul 2026 to Present

Research Intern

Supervision: Prof. Atlas Wang and Dr. Hezhen (Alex) Hu

MAESTRO (first author)

- Developed a bounded agentic framework for controllable, long-form 3D dance authoring. Combined musical-section planning, interval-scoped language edits, measured revision checks and recoverable checkpoints over existing motion generators so creators can revise local motion without losing accepted choreography.
- Evaluated with **12 dance-experienced participants**; received **47 of 72 overall-preference selections (65.3%)** in blind comparisons against LODGE and EDGE.

Georgia Institute of Technology | EIC Lab

Jul 2025 to Feb 2026

Research Intern

Supervision: Prof. Yingyan (Celine) Lin; Prof. Chaojian Li

T3R (first author)

- Developed training-free vision-language-action (VLA) inference methods combining segmentation-guided token pruning and saliency-guided attention refinement to retain task-relevant evidence. Evaluated three backbones across LIBERO, SIMPLER and RoboTwin in simulation.
- On LIBERO-Spatial with OpenVLA-OFT, removed **85% of scene-camera tokens** while achieving **96% task success** (97% unpruned); improved success by up to **7 percentage points** under object-placement shifts.

MIRA-Quant (lab research project)

- Developed a multi-agent iterative-reasoning system for automated LLM quantization, matching expert-tuned NVIDIA Model Optimizer references **within 0.3 perplexity** across 15 configurations spanning five LLMs and three post-training quantization methods.

SELECTED RESEARCH PROJECTS

Greenify: Energy-Aware Neural Architecture Search

Jan 2025 to Apr 2025

Course research project

- Built a differentiable neural architecture search framework with PyTorch and CodeCarbon; reduced measured inference energy by **18% on MNIST** with **less than 0.2% accuracy loss**.

TEACHING EXPERIENCE

UT Austin | Teaching Assistant, CS 311

Aug 2022 to May 2025

- Mentored **more than 400 students** in Discrete Mathematics, including logic, combinatorics, graph theory and algorithm verification.

INDUSTRY EXPERIENCE

Microsoft | Software Engineer

Aug 2025 to Present

- Own backend services for DNS-based threat intelligence, emphasizing service reliability and staged deployment with rollback.

Microsoft | Software Engineering Intern

Jun 2024 to Aug 2024

- Built and evaluated an ML pipeline for detecting potential data exfiltration in DNS queries, including feature engineering and serverless deployment.

Google | STEP Intern

May 2023 to Aug 2023

- Developed Java data pipelines for targeted notification campaigns.

SELECTED COURSEWORK

Graduate: Reinforcement Learning; Deep Learning; Deep Generative Models.

In progress: Optimization; Planning, Search, and Reasoning Under Uncertainty.

Foundations: Linear Algebra; Probability and Statistics; Algorithms and Complexity.

TECHNICAL SKILLS

Programming: Python, C++, Java, SQL.

Research tools: PyTorch, JAX, Hugging Face Transformers, scikit-learn, LIBERO, Git, Linux.